Peeyoush Shah

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Summary

Final-year Computer Science student focused on applied machine learning, deep learning, and production ML systems. Experience building classical ML pipelines, NLP and computer vision models, data workflows, and deployable AI services using Python, SQL, PyTorch, TensorFlow, FastAPI, and Docker.

Technical Strengths

Languages: Python, SQL, Java, JavaScript, C
Machine Learning & Data: Scikit-learn, Pandas, NumPy, Feature Engineering, EDA, Data Preprocessing, Feature Selection, Model Evaluation, Statistical Analysis
Deep Learning & AI: PyTorch, TensorFlow, BERT, Transformers, NLP, CNNs, Computer Vision, Hugging Face
Backend & APIs: FastAPI, Pydantic, REST APIs
Databases: PostgreSQL,MySQL
MLOps & Tools: Docker, GitHub Actions, CI/CD, Git, GitHub, Linux
Generative AI: LLMs, Ollama, OpenAI-compatible APIs
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning

Projects

- **FEVER Verdict – AI-Powered Fact Verification System**

– Designed HCAP (Hierarchical Cross Attention Pooling), a **BERT-based architecture** using sentence- and document-level attention to classify claims as SUPPORTS, REFUTES, or NOT ENOUGH INFO.
– Trained on **145K FEVER claims** using class balancing, weighted loss, and noise-aware training, improving validation accuracy from **56.4% to 72.6%**.
– Implemented a **BM25 retrieval pipeline** over the FEVER Wikipedia corpus and deployed the end-to-end verification system using **FastAPI and Next.js**.

- **CodeInsight – AI Developer Assistant**

– Developed an intelligent code analysis platform supporting **40+ bug pattern detections** across multiple programming languages using rule-based and AST-based static analysis.
– Built a **FastAPI** backend with project-wide analysis, AI-powered code explanations, JWT authentication, and real-time collaboration through WebSockets.
– Implemented quality scoring and automated code improvement suggestions with **Dockerized deployment, GitHub Actions CI, and automated testing**.

- **ECG Arrhythmia Classification – MIT-BIH** *Ongoing Research*

– Developing an ECG arrhythmia classification pipeline on 105K+ heartbeat samples using dual-lead signals and record-level **StratifiedGroupKFold** validation.
– Systematically evaluating **feature engineering, ANOVA feature selection, class-balancing strategies, and multiple ML classifiers** under controlled experiments.
– Current best pipeline combines derivative features, ANOVA selection, Random Oversampling, and **Random Forest**, achieving **0.346 Macro-F1**.

Education

Kalinga Institute of Industrial Technology **Oct 2023 – May 2027 (Expected)**
Bachelor of Technology in Computer Science (CGPA: 9.11) *Bhubaneswar, Odisha*

Certifications

- **AWS Academy Graduate – Cloud Foundations – Training Badge** – AWS Academy, 20 hours, Aug 2026
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI & Stanford Online
- **Advanced Learning Algorithms** – DeepLearning.AI & Stanford Online
- **Unsupervised Learning, Recommenders & Reinforcement Learning** – DeepLearning.AI & Stanford Online

Achievements

- Participated in Himalayan Java National Basketball League; participated in multiple hackathons across India focused on problem solving and rapid prototyping.